

IMMANUVEL M

Aspiring Robotics Engineer

Chennai, Tamil Nadu — +91 9952002158 — m.immanuvel.11@gmail.com

github.com/immanuvel11 — [LinkedIn](#)

Professional Summary

Aspiring Robotics Engineer with hands-on experience in embedded systems, robotics, CAD design, additive manufacturing, and automation. Built functional robotic systems including a 12-DOF quadruped robot, an intelligent health monitoring robot, and an agricultural automation platform. Experienced in embedded programming, electronics integration, rapid prototyping, and multidisciplinary engineering projects.

Education

B.E. Mechatronics Engineering

2024–2028

KCG College of Technology, Chennai (Affiliated to Anna University)

Experience

CAD & Additive Manufacturing Engineer (Part-Time)

EL's Crafting — 1 Year

- Developed parametric CAD models using CadQuery, Blender, and Fusion 360.
- Managed complete 3D printing workflow from slicing to fabrication and printer maintenance.
- Applied rapid prototyping, hardware calibration, and material optimization for functional products.

Technical Skills

Programming	Python, C++, Arduino, HTML, CSS, JavaScript
Embedded	Arduino Mega, ESP32, Bluetooth, Sensor Integration
Robotics	ROS2, OpenCV, Motion Control, Servo Control, System Integration, Electronics Integration
CAD	Fusion 360, CadQuery, Blender, 3D Printing, Rapid Prototyping
Tools	GitHub, Tinkercad

Projects

GENESIS – 12-DOF Quadruped Robot Dog

Individual

- Designed and developed a fully functional 12-DOF quadruped robot using Arduino Mega and MG995/MG996R servos.
- Implemented gait algorithms enabling walking, turning, reverse motion, and dance routines.
- Integrated embedded electronics, power management, and motion control for stable real-world operation.

KIRA – Health Monitoring & Assistance Robot

Team of 4

- Developed an intelligent service robot for temperature monitoring, guided assistance, and user interaction.
- Led software development and contributed to hardware testing, debugging, and system integration.
- Integrated sensors, embedded control, and dual mobile applications.

TIFAN 2026 – Agricultural Automation System

Team Electronics Lead

- Engineered an Arduino Mega-based automation platform integrating conveyor, grippers, sensors, and motion control.
- Led electronics integration, embedded programming, Bluetooth communication, testing, and troubleshooting.
- Developed mobile application support for monitoring and control.

Achievements

- 3rd Place – Acceleration Event, GKDC Go-Kart Design Challenge, Coimbatore.
- Participant – Young Technocrats 3.0 and TIFAN 2026.